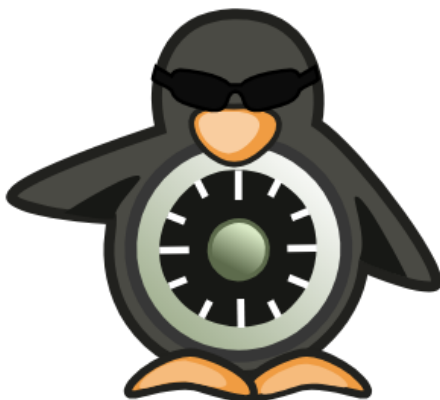


Differences Between SELinux & AppArmor

By: Jason Man, Dominic Yeakel



Summary:

SELinux and AppArmor are both Linux Security Modules (LSMs) that utilize Mandatory Access Control to configure security policies for user accounts, processes, and applications. The following dives into how both operate and the possible use cases.

SELinux:

- Developed by NSA, used by RHEL, Fedora, CentOS, Rocky, and Amazon Linux
- Label-based: security labels are attached to files, processes, ports
- Labels stored as filesystem extended attributes (xattrs)
- Highly precise control
- Policy files written in complex policy language
- Survives file renames/moves since labels follow the inode

```
$ getenforce
Enforcing

$ ls -Z /var/www/html/index.html
system_u:object_r:httpd_sys_content_t:s0 /var/www/html/index.html
```

SELinux: Shows security contexts/labels attached to objects.

AppArmor:

- Uses path-based profiles rather than security labels to restrict program capabilities (comes enabled default on Debian/Ubuntu and openSUSE)
- Easier to write and read policies
- Plaintext files stored in /etc/apparmor.d/ and defines what system files and an application may use
- Focuses on confining individual applications vs. system wide configurations
- Lesser run time overhead, lesser LSM hook crossings

```
$ sudo aa-status
apparmor module is loaded.
12 profiles are loaded.
10 profiles are in enforce mode.

$ ls /etc/apparmor.d/
usr.sbin.apache2
usr.sbin.mysql
usr.sbin.ntpd
```

AppArmor: Shows application profiles defining access rules.

What Determines The Use Case?

1. Operating System
 - a. The linux operating system you are using determines which distro to use
 - i. RHEL/Rocky/Fedora/CentOS/Amazon Linux: SELinux
 - ii. Ubuntu/Debian/openSUSE: AppArmor
2. Compliance
 - a. Governance may force you to use a particular distribution
 - i. For example, an organization may use a RHEL-based environment with SELinux which is commonly used to comply with the PCI DSS framework
3. Accessibility
 - a. Ease of use
 - i. AppArmor is generally easier for administrators to understand and create profiles for individual applications
 - ii. SELinux has a steeper learning curve but offers more detailed policy controls
4. Control Precision
 - a. Controls can be precise or broad
 - i. AppArmor is efficient in restricting applications to files and directories
 - ii. SELinux excels in highly granular control such as with security labels, roles, and domains